

SCI-FLT-MATCHED v0.1

Optimal matched-template map filtering

Historical provisional title pending the explicit r0.5 owner disposition; optimality_status=not_claimed

Scientific Rationale and Contract

STAGE B DRAFT – ALL AO-001 THROUGH AO-006 OPTIONS UNSELECTED.

No numerical route, implementation, conformity result, response or covariance fidelity, observational validation, achieved performance, readiness, production state, scientific freeze, or Unity status is claimed.

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Stage B r0.1 was derived exclusively from the owner-approved eight-object author packet; r0.2 applied the subsequent scientific-owner closure directive; r0.3 applies owner-authorized independent-review repairs without implementation input; r0.4 applies the second independent-review minor repairs and the owner's covariance-scope and diagnostic-only policy dispositions; r0.5 applies the owner's final targeted type, lifecycle, covariance-role, and boundary closure directive while leaving the human title for explicit owner disposition. The admitted packet remains bound by SHA-256

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Part I

Scientific rationale

1 The scientific question

The method answers one narrow question: at each admitted parent-pixel anchor, what is the amplitude of one exact, already-declared template in one exact ordinary-MAP parent? The answer is a map of template amplitude. The method does not decide whether a source exists, choose a candidate, infer a posterior sky, or attach catalog meaning to a peak.

Base v0.1 places one anchor at every exact parent ordinary-MAP pixel center, preserving WCS, frame, grid, shape, indexing, ordering, and the pixel-center convention. It does not interpolate or optimize an anchor. Another output lattice is a separately named successor.

The template is not merely a shape used for smoothing. It is the expected parent-map response per unit amplitude, including its scale, phase, anchor, sampling, support, and calibration/beam lineage. That scaling defines what one unit of the answer means. A template whose unit amplitude denotes point-source flux can support that interpretation only when the exact CAL/BEAM and parent lineages authorize it; another template gives a shape amplitude instead. Exact owner-authorized source-class, morphology, beam, or physical-response authority is admissible. Target payload, target-derived peaks/candidates, filtered output, residuals, NOI members, and post-selection outcomes may not choose, center, tune, or update it.

2 Why the estimator is normalized

The parent is contracted with the template through a frozen complete coordinate-basis weighting object. Every required pixel or quadrature measure is already inside that bilinear object; there is no second unexplained measure. The same weighting is applied to the template itself, producing the normalization. Dividing these two contractions makes an exact matching template return its supplied amplitude on complete valid support. This unit response is the central scientific identity.

Weighting and uncertainty are deliberately not conflated. If the weighting is the exact authorized inverse covariance on the same population, support, conditioning state, and estimable subspace, the estimator has the familiar inverse-covariance GLS minimum-variance property only after a local operator domain is declared before construction, covariance is restricted to that domain, and it is inverted on an exact identifiable estimable subspace. Under a weaker spectral or reliability weight, it can remain matching-template normalized without being minimum variance. In that case the denominator is normalization, not a variance estimate.

3 Why state is explicit and frozen

The transformation must mean the same thing for the signal parent and every NOI realization member admitted by the frozen pre-Apply compatibility predicate. Learn derives candidates and diagnostics from one declared population; Resolve selects the authorized route and freezes its identity; Apply uses it unchanged. Externally declared state enters Resolve. Learning again from each NOI member would change the ensemble method and its uncertainty meaning. New NOI-informed knowledge creates a successor generation rather than rewriting an existing one.

Observation and coadd parents also remain scientifically separate. Filtering an observation and filtering a coadd are two local applications with different parent identities and state generations. This contract neither asserts that filtering commutes with coaddition nor owns a coaddition operation.

4 Why edges are unavailable rather than repaired

The local domain used to construct covariance, weighting, and normalization is declared before the final row exists. Each amplitude then depends on complete application support: every parent payload row with a nonzero coefficient in that row. A coordinate can influence construction while having a final exact zero coefficient; its payload need not be dereferenced, but its causal lineage is retained. Template-query, state-learning, response-query, and storage supports are recorded separately. If one dependency is absent or invalid, the estimator defined here does not exist at that location. Filling, extending, renormalizing, fitting a background, or learning an edge taper would define another method. Numerical fill is harmless only after erosion proves that no admitted output depends on it.

5 Response and uncertainty are full scientific objects

Unit self-response does not make the response a single symmetric beam or make neighboring sources disappear. For a general linear sky, the expectation is the full response-weighted sum of template amplitudes plus the linear action on any residual mean, offset, gradient, or other deterministic background. The off-diagonal response can be anisotropic, asymmetric, nonstationary, position-dependent, or nonlocal. Reducing it to a kernel, radial profile, or solid angle is allowed only after the required invariances, measure, and valid domain are established. Fixed-state response is not the response of a data-dependent Learn–Resolve procedure. That full-procedure quantity requires an authorized rerun or finite difference and is otherwise unavailable.

Likewise, an available conditional covariance companion exists only when its selected scope, matching parent law, pre-draw frozen condition $h_{\text{pre}} = (g_{\text{resolved}}, \theta)$, and fixed finite numerical codomain are authoritative. Apply outcomes and publication facts are provenance, not conditioning variables. Actual finite-precision production is represented by a fixed-state map F_g , not presumed to be linear or guaranteed to return a numerical field on every draw. A bundle always carries each covariance role’s identity and status; an available U1 companion carries the operational covariance on its fixed domain, while typed unavailability need not invalidate the signal. Matrix propagation is used only after fixed-state linearity is established. The separate exact-GLS reference variance $v_{\text{GLS,reference}} = d_p^{-1}$ can remain theorem-supported when an operational covariance companion is unavailable. Exact-reference quantities remain comparators. U2 calibration/template uncertainty, learned-state, numerical discrepancy, representation, NOI sampling, and template-mismatch contributions have different populations and conditioning. The normative uncertainty budget keeps them separate rather than producing an unsupported scalar total.

6 Product and governance rationale

The primary science product is one atomic signal bundle: amplitude map plus the facts required to know exactly what it means. Covariance, empirical NOI, response, and state/lineage products are independently atomic companions. Their failure is explicit. Optional failure does not silently alter the signal estimand, while failure of a required signal member prevents publication of a complete signal

bundle. Each role has its own validity domain no broader than signal validity. An optional unrequested or unavailable role does not erode the signal domain.

The product also preserves a first-class FLT to FRUIT handoff. Its finite v0.1 query vocabulary makes the named FLT-side scientific facts available or exactly reconstructable without exposing undocumented implementation state. A future query outside that vocabulary requires a successor FLT envelope rather than retroactive overpromising. The handoff intentionally says nothing about FRUIT’s future model, iteration, selection, response, uncertainty, or interpretation.

Finally, the contract separates claim layers that are often confused: scientific authority defines the method; conformity concerns a specific implementation; representation fidelity concerns an exact encoding; SCI-VAL evaluates named-use profiles; observational validation and achieved performance are evidence classes; readiness is a governance decision; and production is an operational state. Evidence in one layer does not promote another.

Part II

Shared normative core

The following shared modules are imported unchanged into both the scientist- and engineering-facing documents. They are the single normative source for all definitions, equations, assumptions, requirements, contract consequences, option alternatives, failure rules, and validation consequences.

7 Normative notation and interpretation

7.1 Normative language

“Shall” is a contract requirement. “May” identifies an allowed form whose scientific preconditions still have to be met. “Exact” means equality in the declared mathematical realization. Ordinary finite-precision acceptance is an engineering numerical-conformance question and does not weaken an exact scientific identity. “Unknown” and “unavailable” never mean zero. An option in Section 13 is a decision candidate, not a selected route.

7.2 Collision-free objects, domains, and indexing

The paired readout-coordinate symbols x and r remain reserved to KID authority, and bare Q_p remains reserved to MAP/JINC authority. They are not used for this package’s map-domain anchors or weighting.

A matched-template output anchor is p , a parent-map row or pixel is q , and a separately indexed template anchor is s . The base-v0.1 anchor set $\mathcal{D}_p$ contains exactly one anchor at each parent ordinary-MAP pixel center, with the same WCS, frame, grid, shape, indexing, ordering, and pixel-center convention as the parent apart from exact support restriction. The exact parent row-identity/fact domain is $\mathcal{S}_{\text{parent fact}}$ and its numerical payload domain is $\mathcal{D}_m = \{q \in \mathcal{S}_{\text{parent fact}} : \text{an admitted finite real parent signal payload exists}\}$. The sample m_q exists only for $q \in \mathcal{D}_m$. The expected parent response at row q per unit amplitude anchored at p is t_{pq} ; its complete parent-coordinate vector is t_p^{full} . Before any local inversion or weight construction, the selected route declares the finite local operator domain $\mathcal{D}_{\text{loc}}(p)$ and its exact template extraction

E_p , so $t_p = E_p t_p^{\text{full}}$. The scientific weighting W_p is the complete coordinate-basis bilinear operator on those restricted vectors. Only after W_p , d_p , and the row are constructed is the normalized local covector $\ell_p^* = W_p t_p / d_p$ embedded into parent coordinates as $c_p = E_p^\top \ell_p^*$. Application support is exactly $\mathcal{S}_{\text{apply}}(p) = \{q : c_{pq} \neq 0\}$. Exact zero is the canonical scientific representation; no numerical threshold defines support. The fixed-state row functional is $L_{p,g} u = c_p^\top u$, the template matrix whose column s is t_s^{full} is T , and $R_{\text{fixed}}(p, s) = L_{p,g} t_s^{\text{full}}$.

All scientific input and output vectors are real. A transform representation may be complex only when it preserves real-field conjugate symmetry and uses the Hermitian adjoint. In the real parent-coordinate basis the adjoint is the ordinary transpose. No stationarity, isotropy, periodicity, flat-sky metric, or Euclidean physical metric follows from the vector notation.

Symbol	Normative meaning
p, q, s	Output anchor, parent row/pixel, and independently indexed template anchor.
$\mathcal{S}_{\text{parent fact}}, \mathcal{D}_m$	Exact parent row-identity/fact domain and its subset with admitted finite real signal payload.
m, m_q	Immutable admitted ordinary-MAP parent numerical payload and one sample, defined only on $\mathcal{D}_m$.
t_{pq}, t_p	Parent response per unit amplitude and its complete local-domain vector.
$\mathcal{D}_{\text{loc}}(p)$	Predeclared parent-coordinate domain used to construct the local template, weight, normalization, and operator before final application support is known.
E_p	Exact extraction/restriction from parent coordinates to $\mathcal{D}_{\text{loc}}(p)$.
$\mathcal{S}_{\text{apply}}(p)$	Parent payload coordinates whose final exact $L_{p,g}$ coefficients are nonzero.
ℓ_p^*, c_p	Normalized local covector $W_p t_p / d_p$ and its exact parent-coordinate embedding $E_p^\top \ell_p^*$.
W_p	Frozen, self-adjoint, positive-semidefinite complete bilinear weight; all measure factors required by the scientific contraction are already in it.
n_p	Sparse weighted parent contraction $\sum_{q \in \mathcal{S}_{\text{apply}}(p)} c_{pq} d_p m_q$.
d_p	Weighted template self-contraction $t_p^\top W_p t_p$; exactly real and positive for an admitted anchor.
$\hat{a}_p$	Published fixed-template amplitude n_p / d_p .
$L_{p,g}$	Exact fixed-state scientific row functional producing $\hat{a}_p$.
F_g	Actual fixed-state numerical producing map from admitted parent payloads to the realized amplitude field; it need not be exactly linear.
T	Full template-response operator over the declared template-anchor domain.
$R_{\text{fixed}}(p, s)$	Fixed-state response $L_{p,g} t_s^{\text{full}}$.
$R_{\text{FP}}(p, s)$	Full-procedure response when Learn-Resolve is rerun under an authorized perturbation definition; otherwise unavailable.
$R_{\text{realized}}^{[\delta, \sigma]}(m; p, s)$	Operational fixed-state response of F_g at reference parent m under the declared derivative or finite-difference specification. Here δ is the step (zero for an exact directional derivative) and σ is the derivative/sidedness scheme. A parent-independent matrix shorthand is allowed only after fixed-state linearity is established.
$C_{\text{parent} h_{\text{pre}}}$	Authorized coordinate-basis parent covariance for one exact population conditional only on the pre-draw frozen condition h_{pre} .

Symbol	Normative meaning
C_p	Local-domain covariance $E_p C_{\text{parent} \text{h}_{\text{pre}}} E_p^\top$.
$C_{U1,\text{reference}}$	Theorem/reference covariance $P_C L C_{\text{parent} \text{h}_{\text{pre}}} L^\top P_C^\top$.
$C_{U1,\text{realized}}$	Operational conditional stochastic covariance $\text{Cov}[P_C F_g(m) \mid h_{\text{pre}}]$ on one fixed predeclared finite numerical codomain for the exact U1 population.
P_C	Fixed, predeclared selector from the produced field to the numerical codomain on which a named covariance role is defined; it is independent of the random draw and production outcome.
$\tilde{d}_p$	Optional declared numerical normalization observable exposed by a numerical profile. It exists only when the producing route has a separable denominator with exact identity; it is not a required product of a general F_g .
θ	Frozen template/calibration state on which U1 conditional stochastic statements condition; variation over θ belongs to U2 rather than U1.
g_{resolved}	Immutable method/state generation after Resolve and before the random draw or Apply outcome.
$h_{\text{pre}} = (g_{\text{resolved}}, \theta)$	Pre-draw frozen condition containing only parent/class, template, local-domain/extraction, weight, support and missingness rules, subspace/rank/null, regularization, numerical profile, P_C , response-query domain, and failure semantics. Attempt, realization, publication, outcome, and observed-value facts are provenance, not U1 conditioning variables.
$\mathcal{V}_{\text{signal}}$	Anchors satisfying every predicate required to publish the amplitude signal.
$\mathcal{V}_{\text{response}}(r)$, $\mathcal{V}_{\text{covariance}}(r)$	Signal-valid subdomains on which named response or covariance role r is valid.
$\mathcal{V}_{\text{NOI}}, \mathcal{V}_{\text{FRUIT handoff}}$	Signal-valid subdomain or bundle-level validity for the NOI child and FLT-to-FRUIT handoff roles.
<code>optimality_status</code>	One of <code>established_exact_local_GLS</code> , <code>not_claimed</code> , or <code>unavailable</code> ; title, filename, and method ID are never evidence.
$\mathcal{Q}_{\text{FLT}}^{0.1}$	Finite FLT-owned state/response handoff query vocabulary for this contract version; queries outside it require a successor envelope.

7.3 Coordinate-basis measure and covariance convention

This package uses coordinate-basis column vectors and ordinary matrix multiplication. There is no additional measure-weighted bracket in the scientific estimator. Every fixed pixel area, quadrature weight, transform normalization, or metric factor required by a selected realization is part of W_p (and of the corresponding template/covariance coordinate convention).

For a random parent vector with mean $\bar{m}$,

$$C_{\text{parent}} = \text{E}[(m - \bar{m})(m - \bar{m})^\top], \quad [C_{qq'}] = [m]^2.$$

It is a coordinate-basis covariance matrix, not an integral-kernel symbol. Under an invertible coordinate or quadrature reparameterization $m' = Sm$,

$$C' = S C S^\top, \quad t' = S t, \quad W' = S^{-\top} W S^{-1},$$

so $t^\top W m$ and $t^\top W t$ are invariant. A complex representation replaces transpose by Hermitian adjoint. Thus self-adjointness and positivity mean $W_p = W_p^\dagger$ and $u^\dagger W_p u \geq 0$ in the declared coordinate representation.

The units satisfy $[t] = [m]/[a]$ and $[W] = [m]^{-2}$ for an exact covariance inverse; weaker weights may carry a common nonzero scale and different declared units because that scale cancels in n_p/d_p . In every case

$$[\hat{a}] = [m]/[t].$$

A point-source flux interpretation exists only if the template producer and CAL/BEAM lineage establish that exact amplitude convention. Otherwise the result is a declared shape amplitude.

8 Normative scientific definitions

8.1 Method, estimand, and fit boundary

SCI-FLT-MATCHED is a normalized matched-template map amplitude estimator. The word “optimal” applies only when the exact criterion and premises in Section 9.3 are established. It publishes a filtered amplitude map. At each fixed parent-pixel anchor it performs a fixed-template, fixed-anchor, one-parameter linear amplitude estimate. It does not select candidates, optimize position, fit morphology or background, deblend, infer source existence, or produce a catalog. It is not a posterior/Wiener sky reconstruction. Ordinary convolution does not acquire SCI-FLT-MATCHED identity or amplitude-estimator semantics merely through numerical resemblance. A selected realization may coincide numerically with a normalized correlation or convolution under proved special conditions; such coincidence does not weaken the fixed-template estimand or authorize generic convolution semantics.

Each realization has the immutable scientific identity

```
(method_id, estimand, parent_id, parent_class, anchor_lattice, WCS/frame,
template_id, requested/effective/resolved/applied/realized generation,
weighting_object, exact_operator, producing_map, established_linear_operator,
numerical_profile, response_family, support_family, units/calibration,
covariance_scope, NOI_lifecycle, product_bundle, validity, failure_policy,
optimality_status, provenance).
```

A scientifically consequential change creates a new realization, generation, or separately named method. A representation-only change that exactly preserves the scientific object and supported query vocabulary does not change the estimand or scientific generation.

8.2 Parent classes and anchor lattice

Exactly one parent class is admitted per application: one immutable normalized ordinary-MAP observation bundle, or one immutable normalized ordinary-MAP coadd bundle with its exact contributing-observation membership. They are distinct applications. This contract asserts no filter/coadd commutation and owns no coaddition. JINC and other derived parents are not admitted. Numerical computability cannot repair missing MAP authority.

The base-v0.1 owner disposition places one output anchor at every exact parent ordinary-MAP pixel center. The output amplitude field preserves the parent’s WCS, frame, grid, shape, indexing, order, phase origin, and pixel-center convention, except that exact support rules may make anchors unavailable.

No interpolation is part of anchor creation. Template translation at an anchor, including even/odd sampling and ties, follows the exact deterministic template object. Subpixel, oversampled, and independent output lattices are successors.

8.3 Template-response scientific object

Each application binds one immutable template-response scientific object from exact owner-authorized source-class, morphology, beam, or physical-response authority. It binds the unit-amplitude convention, source-domain reference, parent response, anchor/WCS relation, sampling, phase, support, CAL/BEAM lineage, validity, generation, and supported query vocabulary.

The template shall not be learned, selected, centered, tuned, updated, or reparameterized from target MAP payload, a target-derived candidate or peak, the resulting amplitude field, target residuals, any NOI member, or a post-selection outcome. A predeclared template bank is admissible only when its member-selection rule is fixed independently of target outcomes. Selecting after inspecting target data defines a different inference method.

The same scientific object may be represented by exact materialization, an exact structured form, or exact lineage/query reconstruction. A representation-only change shall preserve every supported query and shall not alter the amplitude estimand. Template mismatch is not silently absorbed into normalization or parent covariance.

8.4 Five support roles

The following supports are distinct:

$\mathcal{S}_{\text{apply}}(p)$.

Exact parent payload rows with nonzero coefficients in $L_{p,g}$. Exact-zero coefficients neither require nor dereference their payloads.

$\mathcal{S}_{\text{template}}(p)$.

Exact rows and metadata required to define or query t_p .

$\mathcal{S}_{\text{state}}(g)$.

Complete population influencing learned covariance, spectra, reliability, window, rank, null, regularization, or other resolved state.

$\mathcal{S}_{\text{response}}(p, s)$.

Exact state, template, and operator support required by the declared response query.

$\mathcal{S}_{\text{store}}$.

Representation footprint. It has no scientific authority and may change losslessly.

The pre-inversion local operator domain $\mathcal{D}_{\text{loc}}(p)$ is declared before E_p , W_p , d_p , or $L_{p,g}$ is constructed. It is not inferred from the final exact-zero pattern. Coordinates in $\mathcal{D}_{\text{loc}}(p) \setminus \mathcal{S}_{\text{apply}}(p)$ may influence local covariance, weight, rank, normalization, or state and therefore remain in operator/state lineage even though Apply does not dereference their parent payload values. Every $q \in \mathcal{S}_{\text{apply}}(p)$ shall lie in $\mathcal{D}_m$ and be admitted. A coordinate in $\mathcal{D}_{\text{loc}}(p) \setminus \mathcal{S}_{\text{apply}}(p)$ may participate in construction without any signal-payload dereference. The shorthand $m_p = E_p m$ is permitted only when a complete numerical local vector exists or when an explicit algebraic completion is proved irrelevant outside $\mathcal{S}_{\text{apply}}(p)$. An output anchor is admitted only when complete application support is present and valid. No partial-support estimator, edge renormalization, extension, imputation, background subtraction, learned taper, or signal-derived support belongs to this method. Numerical fill is allowed only after exact erosion proves zero application coefficient. Causal provenance retains $\mathcal{S}_{\text{state}}$ even when a fixed-state derivative touches only $\mathcal{S}_{\text{apply}}$.

8.5 Weighting and local covariance order

W_p is the selected frozen complete bilinear weight, exactly self-adjoint and positive semidefinite. A parent coefficient has only its selected-route role; its name, positivity, units, reciprocal, or diagonal placement does not make it covariance, precision, or inverse variance.

Local GLS first restricts the authorized parent covariance, conditional on h_{pre} , to the predeclared local domain: $C_p = E_p C_{\text{parent}|h_{\text{pre}}} E_p^T$, then inverts it only on the exact declared local estimable subspace. In general

$$(E_p C_{\text{parent}|h_{\text{pre}}} E_p^T)^{-1} \neq E_p C_{\text{parent}|h_{\text{pre}}}^{-1} E_p^T.$$

A global-precision subblock is generally not the inverse of the locally marginal covariance. Under additional distributional assumptions it may equal a conditional precision, but using that block without the corresponding conditional mean and outside-data terms is not a complete conditional estimator. It is therefore a separately authored weight and method with a different influence domain, response, covariance, and edge rule.

8.6 Learn–Resolve–Apply state graph

State advances through three explicit operations:

Learn.

Derive candidate covariance, spectral/reliability information, window, rank, null, regularization, and diagnostics from one declared population, with source/target protection and state uncertainty recorded.

Resolve.

Select one owner-authorized **SCI-FLT-MATCHED-A0-001** method and bind its parameters, support, scientific identity, numerical-conformance policy, failure disposition, and immutable generation. Externally declared state enters here without observational learning.

Apply.

Use the resolved state unchanged. No target, candidate, source, or NOI-member update occurs during Apply.

Requested, effective, learned-candidate, resolved, applied, complete- publication-candidate, publication-decided, realized, and published identities are distinct even when their values happen to agree. Observation and coadd states are separate. A compatible NOI member receives the identical resolved transformation, support, normalization, and failure rules. Any update creates a successor; per-member relearning is a different future method.

8.7 NOI compatibility predicate

For a signal realization y and a candidate NOI member z , FLT evaluates a frozen admission predicate $K_{\text{NOI}}(z, y)$ before Apply. The predicate uses only declared input facts: parent class and quantity; WCS, grid, shape, indexing, ordering, and pixel-center convention; units and calibration convention; template identity; resolved state generation; local-domain and missingness semantics; NOI population and membership authority; numerical profile; and failure policy. NOI owns population and membership meaning; FLT owns this transformation-compatibility decision. Compatibility never depends on whether a transformed value later succeeds. Missing required payload makes the affected member or anchor incompatible or unavailable; it never adapts support, changes normalization, or triggers member-specific Learn or Resolve.

8.8 Fixed-state, full-procedure, realized, and reference response

The fixed-state response is $R_{\text{fixed}}(p, s) = L_{p, g_0} t_s^{\text{full}}$. It may be asymmetric, anisotropic, nonstationary, position-dependent, or nonlocal. A learned procedure has a separately typed full-procedure response obtained only by an authorized exact rerun or finite-difference definition that reruns Learn and Resolve after perturbation. The state-change record is retained separately. Until such authority exists, R_{FP} is **unavailable**; no FLT, FRUIT, SRC, or other consumer may call R_{fixed} the full-procedure response.

The actual fixed-state numerical producing map is F_g . It may include rounding, branches, exceptional-value handling, or typed attempt failure and is not presumed exactly linear. Its realized response is therefore the declared fixed-state directional derivative or predeclared finite difference of F_g without rerunning Learn or Resolve. The general response is $R_{\text{realized}}^{[\delta, \sigma]}(m; p, s)$, where the response-query domain, reference parent, step δ , and derivative or sidedness scheme σ are fixed before evaluation. Every evaluation entering one derivative or finite difference shall return finite numerical values on the same predeclared output-query domain; otherwise that response query is **unavailable**. Conditioning on successful evaluations, censoring failed evaluations, or changing the domain defines a separately named response experiment. Only after evidence establishes $F_g(m) = \tilde{L}_g m$ on the declared domain may the parent-independent matrix identity $R_{\text{realized}} = \tilde{L}_g T$ be used. The exact scientific comparator is $R_{\text{reference}} = LT$. These are not interchangeable.

Every fixed-state expectation, reference variance, reference covariance, and realized covariance statement conditions only on the pre-draw frozen condition $h_{\text{pre}} = (g_{\text{resolved}}, \theta)$. It contains the immutable parent/class, template, $\mathcal{D}_{\text{loc}}/E_p$, weight, support/missingness, subspace/rank/null, regularization, numerical profile, P_C , response-query domain, and failure semantics. Attempt, realized, publication, outcome, and observed values are recorded as provenance and are never U1 conditioning variables. U2 describes variation over the frozen template/calibration state θ and is excluded from those U1 conditional statements. A named operational covariance role is defined only after a fixed, predeclared selector P_C establishes a finite numerical random-vector codomain. It exists only when $P_C F_g(m)$ is finite almost surely under the declared U1 law. Conditioning on successful production, censoring, pairwise deletion, or a draw-dependent domain is not the same covariance role and requires a separately named population and covariance contract.

8.9 Product roles and atomic lifecycle

The required atomic signal bundle is valid on $\mathcal{V}_{\text{signal}}$ and contains the filtered amplitude field plus its quantity/unit, parent and anchor-lattice identity, template convention, resolved and realized operator/state identity, application validity, actual operational-response reference, uncertainty availability, lifecycle, and complete provenance. Failure of any required member prevents a complete publication candidate.

Every signal bundle carries the identity, request status, validity domain, and lifecycle status of each named covariance role. Independently atomic qualified companions may carry an authorized available covariance scope, frozen-NOI uncertainty, response, or exact state/lineage. Every companion has its own scientific identity, support, population, unit, validity, lifecycle, and named-use policy. Under selected SCI-FLT-MATCHED-A0-003-C, typed operational-covariance unavailability does not make the signal bundle partial unless an exact named-use policy requires that companion. Optional failure does not change the signal estimand. No numerator, denominator, kernel, field-power estimate, response slice, inverse scale, standardized field, or intermediate is a public science product without a complete role and policy.

8.10 Lifecycle vocabulary

State	Meaning
not requested	No request exists. This is not disabled.
requested	A named method/product/use was requested; no admission is yet implied.
effective	Boundary resolution determined that the request applies to this parent and method identity.
learned candidate	Learn produced candidate state that has not been owner-authorized, frozen, or resolved for Apply.
disabled	An applicable requested route was explicitly disabled by its authorized state or policy.
unavailable	A required authority, input, option, support, or scientific premise is absent; no numerical value is asserted.
resolved	Method, state, parameters, support, policy, and failure disposition are immutable for the generation.
applied	The resolved operator was invoked on an admitted parent.
complete publication candidate	Every required member and predicate exists and the bundle is eligible for publication; this is not publication, validation, or readiness.
publication decided	Authorized publication policy accepted or rejected one immutable complete candidate without changing its scientific content.
realized	The declared fixed-state producing map generated immutable values and outcome provenance; realization neither implies publication nor changes pre-draw conditioning.
published	An authorized publication decision exposed the immutable product under its declared identity, role, validity, and lineage.
failed	An authorized attempt began but a required operation or closure check failed.
not produced	Policy intentionally produced no artifact after a valid disposition; it is not zero or unavailable.
superseded	An immutable successor exists under an authorized policy; the predecessor is retained and never rewritten.

8.11 SCI-VAL named-use evaluation

SCI-VAL evaluates an owner-approved named-use profile. It retains four independent axes: request, applicability, eligibility, and realization. FLT owns the producer facts, dependency graph, named-use disposition, prescribed consumer action, and failure semantics. VAL registers and evaluates; it does not filter, publish, select an option, invent a missing fact, or perform observational/scientific validation. Each profile keeps separate verdicts for parent admission (PA), state admission (SA), signal publication (SP), conditional-covariance use (CU), NOI-companion use (NU), response-companion publication/use (RU), and FLT-to-FRUIT producer-envelope use (FH).

Each signal bundle has exact MAP and template boundary identities. NOI and FH use additionally records exactly one boundary-request state: `not_requested`, `inapplicable`, `unavailable`, `compatible`, `requested_pending`, or `realized_child_exists`. A missing or unrequested optional child does not narrow $\mathcal{V}_{\text{signal}}$ or block an unrelated role. A child never mutates its parent signal.

8.12 First-class FLT-to-FRUIT producer envelope

The FLT product contract preserves a first-class interface sufficient for a future FRUIT method to consume the filtered amplitude product without reconstructing undocumented FLT implementation state. The finite v0.1 FLT-owned handoff vocabulary $\mathcal{Q}_{\text{FLT}}^{0.1}$ contains exactly: filtered quantity and per-anchor value/validity/lifecycle; parent and anchor identity/WCS; template identity and amplitude convention; local domain and application support; fixed-state scientific and available realized/reference response identities; method, resolved state, producing-map, and numerical- profile identities; named U1/U2 and companion availability; units/calibration; provenance; and the exact supported state/response query declarations. Selected SCI-FLT-MATCHED-A0-004 representation is required for state reconstruction and selected SCI-FLT-MATCHED-A0-005 representation for response reconstruction. A future FRUIT method requiring a query outside $\mathcal{Q}_{\text{FLT}}^{0.1}$ requires a successor FLT envelope or generation; existing signal products remain unchanged. FLT defines no FRUIT model, recurrence, subtraction/add-back, learning, stopping, restart, selection, response, uncertainty, or interpretation.

The handoff role has its own $\mathcal{V}_{\text{FRUIT handoff}}$, either as a subdomain of $\mathcal{V}_{\text{signal}}$ or as a bundle-level predicate. A failed or unrequested handoff does not narrow signal validity.

8.13 Producer ownership and claim layers

MAP owns the immutable parent identity, support, response, covariance, beam, calibration, WCS, lifecycle, and provenance. The template producer owns the unit-amplitude response object. CAL/BEAM owns calibration and beam authority. FLT owns the exact transformation, resolved state, operational realized response/covariance consequences, products, lifecycle, finite handoff query vocabulary, and one-way producer envelope. NOI owns its realization population and empirical uncertainty meaning. FRUIT owns any future FRUIT science. No SRC boundary is created.

Scientific authority, implementation conformity, representation fidelity, observational/scientific validation, achieved performance, readiness, and production are separate claims. This draft establishes none beyond its own proposed scientific text.

9 Reference estimator, response, and uncertainty

9.1 Authoritative coordinate-basis operator

For every admitted anchor $p \in \mathcal{V}_{\text{signal}}$, first declare $\mathcal{D}_{\text{loc}}(p)$ and its extraction E_p , then construct

$$t_p = E_p t_p^{\text{full}}, \quad d_p = t_p^{\text{T}} W_p t_p, \quad (1)$$

$$\ell_p^* = \frac{W_p t_p}{d_p}, \quad c_p = E_p^{\text{T}} \ell_p^*, \quad (2)$$

$$\mathcal{S}_{\text{apply}}(p) = \{q : c_{pq} \neq 0\}, \quad \hat{a}_p = \sum_{q \in \mathcal{S}_{\text{apply}}(p)} c_{pq} m_q, \quad (3)$$

$$n_p = d_p \hat{a}_p, \quad L_{p,g} u = c_p^{\text{T}} u, \quad (4)$$

$$R_{\text{fixed}}(p, s) = L_{p,g} t_s^{\text{full}}. \quad (5)$$

$W_p = W_p^{\dagger}$ and $u^{\dagger} W_p u \geq 0$. The scientific normalization d_p is finite, exactly real, and strictly positive. Failure of one of these exact predicates is typed unavailability or attempt failure, never amplitude zero. Ordinary floating-point evidence for the exact identities belongs to the preregistered numerical-conformance profile. Every active $q \in \mathcal{S}_{\text{apply}}(p)$ shall be in $\mathcal{D}_m$ and admitted. Coordinates used only

to construct t_p , W_p , d_p , or c_p need no signal-payload dereference when their final coefficient is exact zero. Exact zero is not a tolerance. The shorthand $m_p = E_p m$ may be used only when the complete numerical vector exists or an explicit completion is algebraically irrelevant outside $\mathcal{S}_{\text{apply}}(p)$.

9.2 Matching-template and general-sky relations

For the matching single-template model

$$m = A t_s^{\text{full}} + n, \quad E[n \mid h_{\text{pre}}] = 0,$$

linearity gives

$$E[\hat{a}_p \mid h_{\text{pre}}] = A R_{\text{fixed}}(p, s).$$

At $p = s$, exact normalization yields

$$R_{\text{fixed}}(p, p) = 1, \quad E[\hat{a}_p \mid h_{\text{pre}}] = A.$$

For a general linear sky and deterministic residual,

$$m = \sum_s A_s t_s^{\text{full}} + b + n,$$

the exact fixed-state expectation is

$$E[\hat{a}_p \mid h_{\text{pre}}] = \sum_s R_{\text{fixed}}(p, s) A_s + L_{p,g} b. \quad (6)$$

Unit diagonal response does not make off-diagonal response vanish. Neighboring or nonlocal templates may contribute, and a mean, offset, gradient, or other unmodeled deterministic b may bias the one-parameter amplitude. These terms are response and nuisance leakage, not source attribution or deblending.

9.3 Local restriction and constrained GLS theorem

For an authorized parent covariance, restriction to the predeclared local operator domain precedes inversion:

$$C_p = E_p C_{\text{parent} \mid h_{\text{pre}}} E_p^{\text{T}}. \quad (7)$$

Let the columns of U_p be an exact orthonormal basis, in this declared coordinate representation, for the authorized local row/covector subspace and $P_p = U_p U_p^{\text{T}}$ its representation-specific projector. Define

$$C_{p,E} = U_p^{\text{T}} C_p U_p, \quad W_p = U_p C_{p,E}^{-1} U_p^{\text{T}}, \quad U_p^{\text{T}} t_p \neq 0,$$

where $C_{p,E}$ is positive definite. This is not a claim that a restricted global inverse equals the inverse of the restricted covariance.

The competitor class contains exactly the linear estimates $\ell^{\text{T}} m_p$ on the identical predeclared $\mathcal{D}_{\text{loc}}(p)$, with the identical U_p and fixed state, $\ell = P_p \ell$, and unit response $\ell^{\text{T}} t_p = 1$ to the identical projected template. The matched GLS row is

$$\ell_* = \frac{W_p t_p}{t_p^{\text{T}} W_p t_p}.$$

For every competitor,

$$\text{Var}(\ell^{\text{T}} m_p \mid h_{\text{pre}}) - \text{Var}(\ell_*^{\text{T}} m_p \mid h_{\text{pre}}) = (\ell - \ell_*)^{\text{T}} C_p (\ell - \ell_*) \geq 0,$$

and

$$v_{\text{GLS,reference}}(p) = \text{Var}(\hat{a}_p \mid h_{\text{pre}}) = d_p^{-1}.$$

There are no constraints beyond $\ell \in \text{Range}(U_p)$ and $\ell^\top t_p = 1$ in the v0.1 theorem. This is the sole minimum-conditional-variance and d_p^{-1} variance route within that exact competitor class, not among every singular-covariance linear estimator. Covariance-null directions are outside this theorem unless separately authorized. Deliberately excluded or infinite-variance modes, weighting-null modes, and a projected template with no identifiable component remain distinct. A shared numerical null label or an unconstrained Moore–Penrose operation does not establish the theorem. Under an invertible local coordinate change S , the admissible row/covector subspace transforms as $S^{-\top} U_p$; orthonormality and $P_p = U_p U_p^\top$ are conveniences of the chosen representation, not invariant extra science.

9.4 Fixed-state and full-procedure response

For frozen state g_0 ,

$$R_{\text{fixed}}(p, s) = L_{p, g_0} t_s^{\text{full}}.$$

If state was learned from the parent, a full-procedure response exists only under separately authorized rerun authority. With

$$g_\epsilon = \text{Resolve}(\text{Learn}(m + \epsilon t_s^{\text{full}})),$$

it is the declared limit or predeclared finite difference

$$R_{\text{FP}}(p, s) = \lim_{\epsilon \rightarrow 0} \frac{\hat{a}_p(m + \epsilon t_s^{\text{full}}; g_\epsilon) - \hat{a}_p(m; g_0)}{\epsilon}, \quad (8)$$

or the exact finite- ϵ analogue with its step, sidedness, and domain fixed before evaluation. The state-change record is separate. Without this authority R_{FP} is unavailable.

9.5 Realized and reference response and covariance

Let L be the exact selected scientific operator and let the actual fixed-state numerical producing map be

$$y = F_g(m).$$

It is not presumed linear. For an admitted reference parent m , its realized fixed-state response is the authorized directional derivative, written as the $\delta = 0$ derivative case,

$$R_{\text{realized}}^{[0, \text{D}]}(m; p, s) = \left[\text{DF}_g[m] t_s^{\text{full}} \right]_p, \quad (9)$$

or $R_{\text{realized}}^{[\delta, \sigma]}(m; p, s)$ for a predeclared finite difference with step, sidedness, and output-query domain fixed before evaluation and with no Learn or Resolve rerun. Every required evaluation shall be finite on that same domain; otherwise the response query is unavailable. The U1 realized and reference covariance roles, conditional only on the pre-draw frozen condition $h_{\text{pre}} = (g_{\text{resolved}}, \theta)$ and on the fixed covariance selector P_C , are

$$R_{\text{reference}} = LT, \quad C_{U1, \text{reference}} = P_C L C_{\text{parent} | h_{\text{pre}}} L^\top P_C^\top, \quad (10)$$

$$C_{U1, \text{realized}} = \text{Cov}[P_C F_g(m) \mid h_{\text{pre}}]. \quad (11)$$

The realized covariance exists only when $P_C F_g(m)$ is a fixed-dimensional finite real random vector almost surely under the declared U1 law. Successful- production conditioning, censoring, pairwise

deletion, or draw-dependent domains define separately named populations and covariance roles; none is a fallback for the expression above. Only after evidence establishes $F_g(m) = \tilde{L}_g m$ on the declared domain may one use

$$R_{\text{realized}} = \tilde{L}_g T.$$

$$C_{U1,\text{realized}} = P_C \tilde{L}_g C_{\text{parent|h}_{\text{pre}}} \tilde{L}_g^T P_C^T. \quad (12)$$

An amplitude field generated by F_g shall carry the operational realized pair, never the reference pair as though it described actual values. Either U1 covariance exists only with matching parent-law authority. U2 calibration/template-scale uncertainty is a separate named role. A joint U1+U2 product exists only when its random variables, conditioning, cross-covariances, and pushforward are separately authorized. A diagonal representation does not assert independence, and omitted entries are unknown.

9.6 Reference-variance and operational-covariance compatibility

$v_{\text{GLS,reference}}(p) = d_p^{-1}$ is a theorem-supported marginal reference variance only for SCI-FLT-MATCHED-AO-001-A under every premise of Section 9.3. It is not the same role as an operational covariance companion and survives typed unavailability of that companion.

Weight / covariance choice	Reference marginal variance	Operational U1 covariance
SCI-FLT-MATCHED-AO-001-A + SCI-FLT-MATCHED-AO-003-C	May be available as $v_{\text{GLS,reference}}(p)$	Unavailable; no covariance-based standardization, draw, independence, or covariance-dependent use.
SCI-FLT-MATCHED-AO-001-A + complete/projected SCI-FLT-MATCHED-AO-003	May be available	Separately available on $\mathcal{V}_{\text{covariance}}(r)$ when all operational premises hold.
SCI-FLT-MATCHED-AO-001-C or another non-GLS route	d_p is normalization only; reference variance unavailable	Separately available or unavailable under its own role.
Missing or population-mismatched parent covariance	Unavailable	Unavailable.

9.7 Engineering approximation observables

The following are comparison observables, not owner-selected scientific thresholds:

$$e_d(p) = \frac{|\tilde{d}_p - d_p|}{d_p}, \quad e_a(p) = |R_{\text{realized}}^{[\delta,\sigma]}(m; p, p) - 1|, \quad (13)$$

$$e_R = \|P_{\text{out}}(R_{\text{realized}}^{[\delta,\sigma]} - R_{\text{reference}})P_{\text{anc}}^T\|, \quad e_F(m) = \|P_{\text{out}}(F_g(m) - Lm)\|, \quad (14)$$

$$e_C = \|P_{\text{out}}(C_{U1,\text{realized}} - C_{U1,\text{reference}})P_{\text{out}}^T\|. \quad (15)$$

e_d is available only when the preregistered profile declares $\tilde{d}_p$ as a separable realized numerical normalization with exact identity and provenance. A general F_g need not expose such a denominator; then e_d is typed unavailable and direct field/response observables carry the profile. $\tilde{d}_p$ is never promoted to a public science product. The response projection includes its declared reference-parent domain. A linear-operator discrepancy norm may additionally be used only when the $\tilde{L}_g$ premise above is established.

One preregistered engineering numerical-conformance profile binds norms, scales, domains, projections, coverage, floating-point environment, exact subspace projectors, and acceptance rules before evaluation. No package-wide constant is scientific authority. If an authorized operator intentionally changes scientific response, support, null space, or estimand, it receives a separate method/realization identity and publishes that discrepancy.

9.8 Uncertainty budget and non-combinability

ID	Contribution	Contract treatment
U1	Parent stochastic covariance	Use $C_{U1, \text{realized}} = \text{Cov}[P_C F_g(m) \mid h_{\text{pre}}]$ for the exact authorized population, fixed codomain, and conditioning; matrix propagation requires established fixed-state linearity.
U2	Calibration/template scale	Preserve joint CAL/BEAM and parent-template authority; variation over θ and missing cross-covariance are outside U1 and are unavailable rather than zero.
U3	Learned-state uncertainty	Base products condition on frozen state. Marginalization over Learn-Resolve needs separate authority.
U4	Numerical or intentional operator discrepancy	Ordinary numerical error belongs to the preregistered engineering profile. Intentional scientific discrepancy changes method/realization identity.
U5	Covariance scope/representation	Declare complete, projected, or unavailable scope separately from exact representation.
U6	Frozen-NOI empirical uncertainty	A qualified companion with its own population and finite-sample meaning; it is not automatically parent covariance or physical noise.
U7	Template mismatch/unmodeled systematics	Outside the base matching- template covariance and separately reported.

No scalar total uncertainty combines these terms unless every term, cross-term, conditioning relation, unit, and population is authorized. Absence is unavailable, never zero.

10 Assumptions and validity domain

The unbiasedness, response, optimality, and covariance statements are valid only on the intersection of the following declared predicates. Every product shall identify which predicates were evaluated, by whose authority, and with what immutable evidence.

ID	Assumption or predicate
ASM-001	The parent is exactly one immutable normalized ordinary-MAP observation or coadd bundle with authoritative identity, generation, grouping, WCS/frame, support, units, calibration provenance, and coadd membership when applicable.
ASM-002	Parent, template, physical amplitude, filtered amplitude, and coordinate-basis covariance have declared real codomains. The parent and output anchor lattices have the owner-disposed common WCS/frame/grid/indexing relation. Coordinate or quadrature changes transform m, t, C, W according to Section 7.3; a complex representation preserves conjugacy and uses the Hermitian adjoint.
ASM-003	t_p is the one immutable template-response scientific object for the application, supplied by exact owner-authorized source-class, morphology, beam, or physical-response authority. It expresses expected parent response per unit declared amplitude, with fixed scale, anchor, phase, validity, lineage, and query vocabulary in an exact representation. It is not learned, selected, centered, tuned, updated, or reparameterized from target MAP payload, target-derived peaks/candidates, the output field, target residuals, NOI members, or post-selection outcomes.
ASM-004	$\mathcal{D}_{\text{loc}}(p)$ and E_p are declared before local inversion and operator construction; complete final $\mathcal{S}_{\text{apply}}(p)$ is present and valid, every active row lies in $\mathcal{D}_m$, and construction-only rows with exact-zero c_{pq} need no payload. No admitted output depends on fill, extension, imputation, partial support, edge renormalization, adaptive background, learned taper, or signal-derived support. Template, state, response, and storage support remain distinct.
ASM-005	W_p is the exact frozen, self-adjoint, positive-semidefinite complete coordinate-basis bilinear weight named by the selected SCI-FLT-MATCHED-A0-001 alternative, including population, units, rank, null, regularization, window, support, and edge rules.
ASM-006	d_p is finite, exactly real, and strictly positive.
ASM-007	For amplitude unbiasedness, the admitted noise has zero conditional mean on the complete support given $h_{\text{pre}} = (g_{\text{resolved}}, \theta)$ and does not induce target-dependent changes in the template or operator.
ASM-008	For GLS optimality, C_{parent} is the exact authorized covariance for the population conditional on h_{pre} ; C_p is restricted to the predeclared local domain before inversion; $C_{p,E}$ is positive definite on the exact authorized row/covector subspace; the projected template is identifiable; and competitors share that domain, subspace, fixed state, and unit-response constraint.
ASM-009	For covariance, the parent law, actual fixed-state producing map F_g , operational response, support, units, and pre-draw frozen condition $h_{\text{pre}} = (g_{\text{resolved}}, \theta)$ refer to the same realization and generation. Attempt, realization, publication, outcome, and observed values are provenance rather than conditioning variables. A fixed predeclared P_C supplies one finite numerical covariance codomain, and $P_C F_g(m)$ is finite almost surely under the declared U1 law. Reference quantities and U1/U2 roles remain separate; variation over θ belongs to U2.
ASM-010	State follows the declared Learn–Resolve–Apply graph and is frozen before Apply. It is neither target-updated nor NOI-member relearned.

ID	Assumption or predicate
ASM-011	A candidate NOI member satisfies the frozen pre-Apply K_{NOI} predicate in Section 8.7 and receives the identical state, operator, support, normalization, numerical profile, and failure policy as the signal parent.
ASM-012	A realized numerical operator is assessed under one preregistered engineering numerical-conformance profile. An intentional scientific discrepancy has separate method/realization identity.
ASM-013	Each named covariance role has a separate scope and representation; response and state query vocabularies and representations satisfy selected SCI-FLT-MATCHED-A0-003-SCI-FLT-MATCHED-A0-005 lifecycle rules and the finite $\mathcal{Q}_{\text{FLT}}^{0,1}$ handoff vocabulary.
ASM-014	Template and parent calibration lineages are compatible; any claimed point-source flux meaning or calibration covariance has explicit CAL/BEAM authority.
ASM-015	A method option, named-use policy, and lifecycle route are explicitly selected; there is no automatic selector, fallback, or silent substitution.

10.1 Validity table

Dimension	Admission evidence	Invalid or missing case	Required outcome
Parent identity	Exact bundle identity, class, generation, grouping, WCS/frame, support, unit, quantity, calibration lineage, and coadd membership	Missing, mutable, derived, or mixed parent	unavailable before application; no FLT repair or coaddition.
Application class	Observation or coadd declared independently	Assumed commutation or borrowed state between classes	unavailable; require a distinct realization.
Output anchors	One anchor at each exact parent pixel center with identical parent WCS, frame, grid, shape, index, order, and pixel-center convention	Interpolated, subpixel, oversampled, independently sampled, or ambiguous phase/tie rule	unavailable under v0.1; require a separately named successor.
Template	Exact owner-authorized source-class/morphology/beam/physical-response object per unit amplitude with scale, phase, anchor, sampling, lineage, validity, and query vocabulary	Learned, selected, centered, tuned, updated, or reparameterized from target payload, target peak/candidate, output, residual, NOI member, or post-selection outcome; mismatched chart, ambiguous scale, lossy representation, or missing lineage	unavailable; no inferred template.

Dimension	Admission evidence	Invalid or missing case	Required outcome
Local domain and application support	Predeclared $\mathcal{D}_{\text{loc}}(p)$ and E_p , exact c_p , complete nonzero-coefficient payload closure in $\mathcal{D}_m$, and exact-zero erosion proof without a threshold, separately typed from state, template, response, and storage support	Domain inferred from final zeros, or missing, invalid, filled, extended, imputed, or partially supported dependency	Fail Resolve for an invalid construction; mark affected Apply anchors unavailable ; no partial estimate.
Weighting state	Selected SCI-FLT-MATCHED-A0-001 , population, self-adjoint PSD operator, rank/null, regularization, coefficients, window and edge rules bound at Resolve	Unknown, unselected, incompatible, or changing W_p	Fail closed; no fallback.
Normalization	Finite, exactly real, strictly positive d_p	NaN, infinite, complex, zero, negative, or null-confounded	Location unavailable or attempted bundle failed; never write zero.
Numerical realization	One preregistered engineering conformance profile with the exact comparator, projectors, norms, scales, domain, and acceptance rules	Post-hoc profile, uncovered modes, support/subspace mismatch, or failed rule	failed for that profile; a scientific operator change requires distinct identity.
Response	Fixed, full-procedure, realized, and reference types kept distinct, with selected exact domain/query/representation contract	Undocumented kernel, assumed invariance, strengthened fixed-state meaning, or unreconstructable response	Response-dependent product or handoff unavailable .
Units and calibration	$[\hat{a}] = [m]/[t]$ and compatible joint lineage	Unjustified flux, beam, solid angle, DC, extended-source, or covariance meaning	Forbid the claim; required named use is unavailable .
Covariance	Matching authorized parent law, actual F_g , pre-draw frozen condition h_{pre} , fixed predeclared P_C , finite numerical random-vector codomain, named U1/U2 role, scientific scope, and exact selected representation	Missing population or conditioning, failed or nonfinite draws on the fixed domain, selection on successful draws, censoring, pairwise deletion, draw-dependent domains, unproved linear propagation, unnamed calibration terms, omitted correlations treated as zero, or d_p^{-1} used without constrained local GLS	Covariance role unavailable or companion failed; signal may remain complete publication candidate.

Dimension	Admission evidence	Invalid or missing case	Required outcome
NOI parity	Pre-Apply K_{NOI} compatibility and identical frozen transformation	Per-member relearning, post-result admission, adapted support, or mixed generation	Member/anchor or NOI companion unavailable or failed; separate method required.
Lifecycle	Atomic membership, named-use policy, and immutable state	Mixed status, partial signal bundle, silent mutation, or overwritten predecessor	Bundle failed; retain predecessor and successor identities.
FLT to FRUIT	Finite $\mathcal{Q}_{\text{FLT}}^{0.1}$, selected SCI-FLT-MATCHED-A0-004 for state reconstruction where required, selected SCI-FLT-MATCHED-A0-005 for response reconstruction, and documented lineage	Requires an unsupported future query, undocumented implementation state, or supplies FRUIT science	Handoff/query unavailable; require a successor envelope for new queries; signal status is unchanged.

The signal domain $\mathcal{V}_{\text{signal}}$ is the intersection of the predicates required for the amplitude role. For each response or covariance role r , $\mathcal{V}_{\text{response}}(r) \subseteq \mathcal{V}_{\text{signal}}$ and $\mathcal{V}_{\text{covariance}}(r) \subseteq \mathcal{V}_{\text{signal}}$ contain only anchors where that companion's additional predicates hold. Likewise $\mathcal{V}_{\text{NOI}} \subseteq \mathcal{V}_{\text{signal}}$ and the subdomain or bundle-level $\mathcal{V}_{\text{FRUIT handoff}}$ is no broader than signal validity. Bundle-level validity additionally requires the role's own atomic identity and lifecycle members. An optional companion that is not requested or is unavailable never narrows $\mathcal{V}_{\text{signal}}$; consumer-specific validity may be narrower.

11 Normative requirements

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-001	The method identity shall be SCI-FLT-MATCHED. Human title and optimality claim are separate: the owner shall select the title disposition, and every public boundary shall carry <code>optimality_status=established_exact_local_GLS</code> , <code>not_claimed</code> , or <code>unavailable</code> . Title, filename, and method ID are never evidence.
SCI-FLT-MATCHED-REQ-002	The sole signal estimand shall be the fixed-template amplitude at each admitted anchor; the public signal role is a filtered amplitude map.
SCI-FLT-MATCHED-REQ-003	Each application shall bind exactly one immutable normalized ordinary-MAP observation parent or one immutable normalized ordinary-MAP coadd parent. JINC and other derived parents are excluded.
SCI-FLT-MATCHED-REQ-004	Observation and coadd applications shall remain distinct; no commutation, equivalence, state sharing, or FLT-owned coaddition is asserted.

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-005	Each application shall bind one immutable, exactly owner-authorized source-class/morphology/beam/physical-response template object per unit amplitude. Exact materialized, structured, and lineage/query representations are admissible only when scientifically equivalent.
SCI-FLT-MATCHED-REQ-006	The template shall not be learned, selected, centered, tuned, updated, or reparameterized from target MAP payload, a target-derived candidate/peak, the resulting amplitude field, target residuals, any NOI member, or post-selection outcomes. A predeclared bank is admissible only when selection is independent of target outcomes.
SCI-FLT-MATCHED-REQ-007	A realization shall bind collision-free $p/q/s/W_p$ notation, coordinate-basis algebra, $\mathcal{S}_{\text{parent fact}}$, $\mathcal{D}_m$, $\mathcal{D}_{\text{loc}}$, real codomains, WCS/frame, indexing, units, support families, boundaries, estimable subspace, rank/null, regularization, validity, and any complex-transform adjoint convention.
SCI-FLT-MATCHED-REQ-008	The authoritative estimator shall construct $t_p = E_p t_p^{\text{full}}$, $d_p = t_p^T W_p t_p$, $\ell_p^* = W_p t_p / d_p$, $c_p = E_p^T \ell_p^*$, and shall apply only $\hat{a}_p = \sum_{q: c_{pq} \neq 0} c_{pq} m_q$, with every scientific measure factor inside W_p .
SCI-FLT-MATCHED-REQ-009	Every admitted W_p shall be exactly self-adjoint and positive semidefinite and every admitted d_p finite, exactly real, and strictly positive. Numerical tolerances do not alter these scientific predicates.
SCI-FLT-MATCHED-REQ-010	$\mathcal{D}_{\text{loc}}(p)$ and E_p shall be fixed before weight or operator construction; complete final $\mathcal{S}_{\text{apply}}(p)$ shall be required. Every active row shall lie in $\mathcal{D}_m$. Construction-only rows with exact-zero final coefficient shall not require payload dereference; no threshold may define exact zero. Partial support, renormalization, extension, imputation, background subtraction, learned taper, and signal-derived support are excluded.
SCI-FLT-MATCHED-REQ-011	Fixed-state, full-procedure, realized, and reference response shall remain separately typed. Required response queries shall be exactly available or reconstructable; a reduced kernel requires proved invariances and measure.
SCI-FLT-MATCHED-REQ-012	<code>established_exact_local_GLS</code> , the word “optimal,” and $v_{\text{GLS,reference}}(p) = d_p^{-1}$ require the constrained local inverse-covariance GLS theorem: restriction before inversion, a positive-definite estimable covariance, identifiable projected template, identical local domain/subspace/state, the sole unit-response constraint, and the exact competitor class.
SCI-FLT-MATCHED-REQ-013	Every realization shall identify one owner-selected SCI-FLT-MATCHED-A0-001 weighting alternative separately for its parent class. No option in this draft is a default.
SCI-FLT-MATCHED-REQ-014	A parent coefficient shall retain only its declared route-specific role; its name, sign, units, reciprocal, or diagonal form does not imply covariance, precision, inverse variance, or independence.

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-015	State shall follow Learn–Resolve–Apply. Not requested, requested, effective, disabled, unavailable, learned candidate, resolved, applied, complete publication candidate, publication decided, realized, published, failed, not produced, and superseded states remain distinct; Apply uses immutable state and performs no target/member update.
SCI-FLT-MATCHED-REQ-016	Observation and coadd state generations shall remain separate and shall not be substituted for one another.
SCI-FLT-MATCHED-REQ-017	Every NOI member satisfying the frozen pre-Apply K_{NOI} predicate shall receive the identical resolved state, transformation, application support, normalization, numerical profile, and failure rules as the corresponding signal realization.
SCI-FLT-MATCHED-REQ-018	An NOI-informed update shall create an immutable successor; per-member relearning is a different, unavailable-in-v0.1 method.
SCI-FLT-MATCHED-REQ-019	The method shall have no automatic selector or fallback. A requested unavailable route fails closed without silent substitution.
SCI-FLT-MATCHED-REQ-020	Every produced signal bundle shall carry the identity, request status, validity domain, and lifecycle status of each named covariance role. When SCI-FLT-MATCHED-A0-003 selects an available scope and its required parent law and fixed-domain authority exist, the corresponding companion shall carry $C_{U1,\text{realized}} = \text{Cov}[P_C F_g(m) h_{\text{pre}}]$; $C_{U1,\text{reference}} = P_C L C_{\text{parent} h_{\text{pre}}} L^T P_C^T$ remains a separate comparator. Matrix propagation requires established fixed-state linearity.
SCI-FLT-MATCHED-REQ-021	$v_{\text{GLS},\text{reference}}(p) = d_p^{-1}$ shall be called marginal reference conditional variance only under every constrained local GLS premise; otherwise d_p is normalization. Operational-covariance unavailability does not erase this separately supported theorem result.
SCI-FLT-MATCHED-REQ-022	Uncertainty availability and U1–U7 shall be truthful and role-separated. U1 conditional stochastic covariance, U2 calibration/template uncertainty, and any authorized joint product shall have distinct identities; all U1 statements condition only on $h_{\text{pre}} = (g_{\text{resolved}}, \theta)$ and variation over θ belongs to U2; missing covariance, cross-covariance, or entries are unavailable rather than zero or independent.
SCI-FLT-MATCHED-REQ-023	The public signal product shall be the role-complete atomic bundle defined in Section 8.9, valid on $\mathcal{V}_{\text{signal}}$.
SCI-FLT-MATCHED-REQ-024	Failure of any required signal member shall prevent a complete publication candidate; a partial bundle shall not be published as complete.
SCI-FLT-MATCHED-REQ-025	Covariance, frozen-NOI uncertainty, response, and state/lineage companions shall be independently atomic and qualified by exact identity, population, unit, support, their own subdomain or bundle validity, lifecycle, request status, and named-use policy. Optional unavailability shall not narrow $\mathcal{V}_{\text{signal}}$.

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-026	The lifecycle states in Section 8.10 shall remain distinct with the exact definitions in Section 8.10; in particular not requested is not disabled, realized is not published, and a complete publication candidate is not publication, validation, or readiness.
SCI-FLT-MATCHED-REQ-027	No numerator, denominator, kernel, field-power estimate, response slice, inverse scale, standardized field, or intermediate is public without a complete science-product role and policy.
SCI-FLT-MATCHED-REQ-028	The product contract shall preserve the finite first-class $\mathcal{Q}_{\text{FLT}}^{0.1}$ producer envelope in Section 8. A future FRUIT requirement outside that vocabulary requires a successor FLT envelope; it cannot retroactively broaden an existing product.
SCI-FLT-MATCHED-REQ-029	No FLT artifact or profile shall define FRUIT source modeling, subtraction/add-back, recurrence, learning, stopping, restart, selection, response, uncertainty, or interpretation.
SCI-FLT-MATCHED-REQ-030	The method performs fixed-template, fixed-anchor, one-parameter linear amplitude estimation. Candidate selection, position/morphology/ background fitting, deblending, existence inference, catalogs, significance, completeness, purity, and SRC ownership remain outside scope.
SCI-FLT-MATCHED-REQ-031	Posterior/Wiener sky reconstruction, generic ordinary-convolution semantics, data-thresholded destriping, adaptive edge/background methods, and the historical high-pass/delta case remain separate methods. Proved numerical coincidence with normalized correlation/convolution does not change the SCI-FLT-MATCHED estimand. SCI-FLT-FIXED remains protected.
SCI-FLT-MATCHED-REQ-032	Every affected route shall receive its exact scientific-owner, engineering preregistration/representation, or successor-authorship disposition before dependent use is available. Representation choice cannot select science, and no disposition can manufacture an unavailable parent, template, or authority.
SCI-FLT-MATCHED-REQ-033	The scientific method is the exact selected n_p/d_p operator. Ordinary finite precision is evaluated under one preregistered engineering numerical-conformance profile. A scientifically different operator receives a separate method/realization identity and retains its discrepancy.
SCI-FLT-MATCHED-REQ-034	Scientific authority, implementation conformity, representation fidelity, observational validation, achieved performance, readiness, and production shall be separate claims with separate evidence and owners.
SCI-FLT-MATCHED-REQ-035	SCI-VAL shall evaluate an owner-approved named-use profile on separate request, applicability, eligibility, and realization axes. It shall not perform filtering, publication, or observational/scientific validation.
SCI-FLT-MATCHED-REQ-036	Parent and template calibration dependence shall remain joint. Point-source flux, matched-filter beam/solid angle, extended-source fidelity, and calibration covariance require exact separate authority.

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-037	A coadd application shall retain exact contributing-observation membership and parent provenance without asserting filter/coadd commutation.
SCI-FLT-MATCHED-REQ-038	Every signal product shall bind exact MAP and template boundary identities. NOI and FLT-to-FRUIT records shall carry exact identity and one named-use request state— <code>not_requested</code> , <code>inapplicable</code> , <code>unavailable</code> , <code>compatible</code> , <code>requested_pending</code> , or <code>realized_child_exists</code> —including K_{NOI} and $Q_{\text{FLT}}^{0.1}$ where applicable; the full realization tuple; immutable provenance; supported query/reconstruction contract; and a route-status record.
SCI-FLT-MATCHED-REQ-039	This Stage B draft and every unselected option shall make no implementation, conformity, covariance/response fidelity, observational validation, performance, readiness, production, scientific-freeze, or Unity claim.
SCI-FLT-MATCHED-REQ-040	Base v0.1 shall evaluate one anchor at each exact parent pixel center and preserve parent WCS/frame/grid/shape/index/order/pixel-center conventions. No interpolation is admitted; other lattices are successors.
SCI-FLT-MATCHED-REQ-041	Local covariance shall be $C_p = E_p C_{\text{parent} \text{hpre}} E_p^T$ on the predeclared $\mathcal{D}_{\text{loc}}(p)$, and local GLS shall invert only its exact authorized subspace. A global-precision subblock is neither that marginal inverse nor a complete conditional estimator without its outside-data and conditional-mean terms.
SCI-FLT-MATCHED-REQ-042	$\mathcal{S}_{\text{parent fact}}$, $\mathcal{D}_m$, $\mathcal{D}_{\text{loc}}$, $\mathcal{S}_{\text{apply}}$, $\mathcal{S}_{\text{template}}$, $\mathcal{S}_{\text{state}}$, $\mathcal{S}_{\text{response}}$, and $\mathcal{S}_{\text{store}}$ shall be separately recorded. Final exact-zero application coefficients do not dereference parent payload but retain causal operator/state lineage when they influenced construction.
SCI-FLT-MATCHED-REQ-043	The general-sky expectation shall retain all off-diagonal template response and $L_{p,g}b$ nuisance leakage. Unit diagonal response shall not imply source attribution, isolation, or absence of bias.
SCI-FLT-MATCHED-REQ-044	When state is parent-learned, full-procedure response requires an authorized Learn–Resolve rerun or predeclared finite difference and a separate state-change record; otherwise it is unavailable.
SCI-FLT-MATCHED-REQ-045	Any amplitude field generated by F_g shall carry its operational fixed-state realized response definition $R_{\text{realized}}^{[\delta,\sigma]}(m; p, s)$ and any authorized $C_{U1,\text{realized}}$, not reference quantities mislabeled as actual. Response finite differences and covariance shall use fixed predeclared finite numerical domains with no success conditioning, censoring, pairwise deletion, or draw/evaluation-dependent domain. $\tilde{L}_g T$ and matrix covariance propagation are admissible only when $F_g = \tilde{L}_g m$ is established on the declared domain.

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-046	For each named covariance role, scientific scope shall be selected independently as complete within that role, named projected, or unavailable; exact representation shall be selected independently and shall not change the role or covariance identity. SCI-FLT-MATCHED-A0-003-C shall carry typed role identity and unavailability and shall not make the signal bundle partial unless an exact named-use policy requires that companion. It denies operational covariance and covariance-derived uses, not a separately theorem-supported $v_{\text{GLS}, \text{reference}}$ under SCI-FLT-MATCHED-A0-001-A .
SCI-FLT-MATCHED-REQ-047	Exact immutable state, its finite required query vocabulary, and its own role validity shall be reproducible across full, compact, or lineage representation without changing the estimator or signal generation. Representation is an engineering selection after the scientific vocabulary is fixed.
SCI-FLT-MATCHED-REQ-048	Response domain, finite query vocabulary, $\mathcal{V}_{\text{response}}(r)$, and consumer scope shall be defined before engineering chooses materialized, structured, or lineage representation. $\mathcal{Q}_{\text{FLT}}^{0.1}$ is the complete current handoff vocabulary; FLT shall not guess future FRUIT requirements.
SCI-FLT-MATCHED-REQ-049	PA, SA, SP, CU, NU, RU, and FH verdict meanings, separately addressable request/status records, role validity, and dependency graph are normative. An optional unrequested role shall not block unrelated SP. Profile count/layer/vector layout is lossless representation only; a scalar verdict shall not drive scientific action.
SCI-FLT-MATCHED-REQ-050	Covariance zero-variance modes, excluded or infinite-variance modes, weighting-null modes, and template-unidentifiable modes shall remain distinct exact subspace classifications.

12 Falsifiable contract consequences

Stable **SCI-FLT-MATCHED-PRED-** identities cover three explicitly typed classes: analytic scientific consequences, stochastic identities, and lifecycle/conformance invariants. Each is conditional on the cited assumptions and is falsified by a conforming counterexample in its declared domain. Empirical estimators and finite-sample acceptance belong to a preregistered engineering protocol. Failure authorizes no fallback.

The analytic class is **SCI-FLT-MATCHED-PRED-001**, **SCI-FLT-MATCHED-PRED-002**, **SCI-FLT-MATCHED-PRED-014**, and **SCI-FLT-MATCHED-PRED-019–SCI-FLT-MATCHED-PRED-022**. The stochastic class is **SCI-FLT-MATCHED-PRED-003**, **SCI-FLT-MATCHED-PRED-008**, and **SCI-FLT-MATCHED-PRED-024**. Every remaining **SCI-FLT-MATCHED-PRED-** identity, including appended **SCI-FLT-MATCHED-PRED-025**, is a lifecycle/conformance invariant. These type assignments are normative and do not change the stable IDs.

Consequence ID	Falsifiable statement	Assumptions
SCI-FLT-MATCHED-PRED-001	An exact unit-amplitude matching template at admitted p gives $R_{\text{fixed}}(p, p) = 1$.	003–006

Consequence ID	Falsifiable statement	Assumptions
SCI-FLT-MATCHED-PRED-002	At fixed state, a noiseless matching template scaled by A gives exactly $\hat{a}_p = A$.	003–006
SCI-FLT-MATCHED-PRED-003	With zero conditional-mean noise and a matching template, $E[\hat{a}_p h_{\text{pre}}] = A$.	003–007
SCI-FLT-MATCHED-PRED-004	An anchor missing any nonzero application-support dependency is never published as a numerical amplitude.	004,006
SCI-FLT-MATCHED-PRED-005	Nonfinite, nonreal, zero, or negative d_p never produces zero amplitude as a substitute.	005–006
SCI-FLT-MATCHED-PRED-006	Observation and coadd applications retain distinct parent and state identities.	001,010
SCI-FLT-MATCHED-PRED-007	Every NOI member admitted by the frozen K_{NOI} predicate has the identical fixed L , support, normalization, profile, and failure decisions as its signal realization.	010–011
SCI-FLT-MATCHED-PRED-008	For the exact scientific fixed-state linear map and authorized parent law, $\text{Cov}[P_C L m h_{\text{pre}}] = P_C L C_{\text{parent} h_{\text{pre}}} L^T P_C^T$. An NOI ensemble targets this identity only when separate authority establishes the same population, conditioning, fixed codomain, and covariance role.	001, 002, 004, 009, 013
SCI-FLT-MATCHED-PRED-009	Without exact constrained local GLS premises, removing a d_p^{-1} reference-variance label changes no amplitude value and removes the unsupported uncertainty claim. With those premises, typed operational-covariance unavailability does not remove the separately supported reference variance.	005,008–009
SCI-FLT-MATCHED-PRED-010	Acceptance under a preregistered engineering numerical profile implies only that every predeclared profile rule was satisfied; it is a conformance invariant, not a scientific error budget or performance claim.	012–013
SCI-FLT-MATCHED-PRED-011	A consequential realization-field change creates a new identity and never mutates an existing product.	010, 013, 015
SCI-FLT-MATCHED-PRED-012	Failure of one required signal member prevents a complete publication candidate.	015
SCI-FLT-MATCHED-PRED-013	Failure of an optional companion leaves a complete signal bundle unchanged unless its named-use policy requires that companion.	009, 013, 015
SCI-FLT-MATCHED-PRED-014	For a nonzero real pure amplitude-coordinate reparameterization $t \mapsto ct$ that holds $\mathcal{D}_{\text{loc}}, E_p, W_p, g$, support, and all non-template state fixed, $\hat{a} \mapsto \hat{a}/c$ while $\hat{a}t$ is invariant. Any re-Learn or re-Resolve operation is a new realization outside this consequence.	003–006,014
SCI-FLT-MATCHED-PRED-015	A one-kernel response representation fails where its required invariances do not hold.	002,013
SCI-FLT-MATCHED-PRED-016	Requesting an unavailable route never silently selects another route.	015

Consequence ID	Falsifiable statement	Assumptions
SCI-FLT-MATCHED-PRED-017	A signal bundle plus its declared response/state interface answers every query in $\mathcal{Q}_{\text{FLT}}^{0.1}$ without undocumented state and returns typed unavailability outside it.	013,015
SCI-FLT-MATCHED-PRED-018	No conforming output alone is a detection, catalog entry, posterior sky, FRUIT result, performance claim, readiness decision, or production decision.	001–015
SCI-FLT-MATCHED-PRED-019	With two nonzero template amplitudes, the expectation at p equals $R(p, s_1)A_{s_1} + R(p, s_2)A_{s_2}$ plus any nuisance term.	003–007
SCI-FLT-MATCHED-PRED-020	A nonzero constant or gradient background changes $E[\hat{a}_p]$ by exactly $L_{p,g}b$ when that contraction is nonzero.	003–007
SCI-FLT-MATCHED-PRED-021	A nonzero off-diagonal $R_{\text{fixed}}(p, s)$ produces a response at p to a unit template at $s \neq p$ despite exact unit diagonal response.	003–006
SCI-FLT-MATCHED-PRED-022	A mismatched signal shape returns its response-weighted template amplitude and need not equal any unconstrained source property.	003–006
SCI-FLT-MATCHED-PRED-023	For parent-learned state, an authorized full-procedure response can differ from fixed-state response exactly through the recorded state change; without rerun authority the former remains unavailable.	010
SCI-FLT-MATCHED-PRED-024	The operational realized U1 covariance is $\text{Cov}[P_C F_g(m) \mid h_{\text{pre}}]$ on its fixed finite codomain, not the reference covariance. If fixed-state linearity $F_g(m) = \tilde{L}_g m$ is established, this equals the matrix expression in Equation 12.	001, 002, 009, 013
SCI-FLT-MATCHED-PRED-025	If a coordinate in $\mathcal{D}_{\text{loc}}(p)$ lacks numerical parent payload but has exact-zero final coefficient c_{pq} , the amplitude remains defined and unchanged without dereference. If the coefficient is nonzero, that coordinate enters $\mathcal{S}_{\text{apply}}(p)$ and the anchor is unavailable.	004–006

13 Unselected authored option sets

All alternatives below remain unselected. Every scientific weight is exact, self-adjoint, positive semidefinite, and used through exact normalization. Observation and coadd dispositions and generations remain separate.

13.1 SCI-FLT-MATCHED-A0-001: Scientific weighting authority

Each route binds the map chart, units, support, population, Learn/Resolve identity, exact subspaces, window/boundary action, regularization, coefficient roles, diagnostics, state uncertainty, generations, and failure disposition.

13.1.1 SCI-FLT-MATCHED-A0-001-A: exact constrained local inverse-covariance GLS

Science.

Declare $\mathcal{D}_{\text{loc}}(p)$ and E_p , restrict exact authorized C_{parent} to that domain, then use $W_p = U_p C_{p,E}^{-1} U_p^T$ on the declared estimable subspace with identifiable projected template. This alone supports the theorem in Section 9.3.

Failure and evidence.

Missing covariance authority, population or conditioning mismatch, nonpositive $C_{p,E}$, subspace mismatch, or unidentifiable template makes GLS/optimality unavailable. Future assessment checks the exact theorem and the preregistered numerical profile separately.

13.1.2 SCI-FLT-MATCHED-A0-001-B: structured covariance-derived weighting

Science.

This is an authorship envelope, not yet a selectable weight. A decision-ready successor must supply one concrete exact self-adjoint PSD W_p derived from a declared covariance family. Once authored, that W_p is exact science even when it is not the exact covariance inverse. Normalization and unit response remain; optimality and d_p^{-1} variance do not.

Failure and evidence.

Unbound structure, population, modes, support, regularization, or covariance ancestry fails the route. Any comparison with an inverse covariance is an engineering/profile observable, not an implied claim.

13.1.3 SCI-FLT-MATCHED-A0-001-C: radially symmetrized field-power spectral weighting

Identity.

If owner-authorized, the separately named nonoptimal method is `radially_symmetrized_field_power_spectral_weighting` and carries `optimality_status=not_claimed`.

Object.

For each predeclared $\mathcal{D}_{\text{loc}}(p)$, let ξ_q be the declared local-chart coordinate and define the exact analysis operator

$$(A_p u)_k = \alpha_p \sum_{q \in \mathcal{D}_{\text{loc}}(p)} H_p(q) u_q e^{-2\pi i k \cdot \xi_q} \mu_q, \quad F_{jp}(k) = (A_p E_p f_j)_k,$$

for a declared field population f_j . Define

$$B_{pb} = \{k : \rho_p(k) \in [\rho_{pb}, \rho_{p,b+1})\}, \quad \overline{P}_{pb} = \frac{\sum_{j,k \in B_{pb}} w_{pjk} |F_{jp}(k)|^2}{\sum_{j,k \in B_{pb}} w_{pjk}}.$$

where $\rho_p(k)$ is the exact radial coordinate under the declared local chart and WCS metric. The source may be an independently authorized noise population or a declared parent-derived field. Its name and deterministic-residual imprint are retained.

Exact conventions.

Each local radial bin B_{pb} is half-open in the declared radial coordinate except the declared final-boundary rule; boundary ties use one deterministic assignment. Member/mode weights are finite and nonnegative and their denominator is positive. Phase, normalization, units, conjugate modes, multiplicities, averaging order, WCS metric, window, and support are exact.

Weight and modes.

Let D_p be diagonal on a conjugate-complete mode set, with

$$(D_p)_{kk} = w_{\text{spec},p}(k) = [\bar{P}_{p,b(k)} + \lambda_{p,b(k)}]^{-1}$$

on admitted modes and zero action on explicitly excluded modes, where $\lambda_{pb} \geq 0$ has the units of $\bar{P}_{pb}$. The exact coordinate-basis weight is

$$W_p = A_p^\dagger D_p A_p.$$

The selected phase, conjugacy, multiplicity, and real-coordinate restriction shall make W_p real, self-adjoint, and positive semidefinite and shall declare its units and final support. Regularized modes and excluded-null modes are distinct. A sparse bin is never interpolated or borrowed from neighbors without a separately authored rule.

Consequences.

Radialization deliberately loses angular information and implies no noise, covariance, stationarity, isotropy, or optimality claim. d_p is normalization. Matching covariance, if any, requires independent authority.

Failure and evidence.

Invalid weights/denominator, nonfinite or negative power, unbound ties/multiplicity, missing field provenance, incomplete support, or hidden borrowing fails the route. Future evidence reports bin membership, source imprint, anisotropy/dispersion, leakage, regularization, response, and NOI parity as mandatory diagnostics without promoting them to scientific thresholds. Their exact diagnostic estimators and normalizations are preregistered engineering evidence fields and never determine v0.1 admission.

13.1.4 SCI-FLT-MATCHED-A0-001-D: declared weaker positive-semidefinite weighting

Science.

This is a successor-authorship trigger, not a selectable catch-all. A future candidate must supply one concrete exact self-adjoint PSD map- or transform-domain W_p with declared purpose, population, units, support, rank/null, regularization, and lineage, but no covariance ancestry claim.

Failure and evidence.

Non-self-adjointness, indefiniteness, hidden coefficients, state drift, incomplete support, or invalid d_p fails the route. No optimality or variance meaning follows.

13.2 SCI-FLT-MATCHED-A0-002: Operator realization and numerical-conformance class

No numerical constant in this family is scientific authority. Exactly one engineering profile is preregistered per realized numerical identity.

13.2.1 SCI-FLT-MATCHED-A0-002-A: numerical realization of the exact operator

The scientific operator, response, support, subspaces, and failure semantics are exact. The preregistered engineering profile binds floating-point format, algorithms, environment, exact comparator, domains, norms, scales, projectors, coverage, and acceptance rules before results are seen. Profile pass is only a numerical-conformance statement.

13.2.2 SCI-FLT-MATCHED-A0-002-B: intentionally distinct scientific operator

This label is a nonselectable successor-authorship trigger. A concrete route is admissible only through separate owner authority and authorship that states the affected quantity, estimand, response/support/null discrepancy, scientific error budget, norm/domain, calibration consequence, consumers, and rationale. It creates a distinct method or realization identity; it is not ordinary finite precision and cannot borrow the exact operator's response/covariance.

13.2.3 SCI-FLT-MATCHED-A0-002-C: numerical route unavailable

If no preregistered profile, exact comparator, domain/coverage, or required projector exists, no numerical-conformance claim is available. No threshold, iteration count, percentile, or post-hoc test supplies a fallback.

13.3 SCI-FLT-MATCHED-A0-003: Conditional-covariance scope and representation

This family requires one scientific-scope disposition from A–C and, for an available scope, one exact-representation disposition from D–E. Equivalent exact representations retain one covariance identity. Every signal bundle retains the identity and lifecycle status of each named covariance role, including an unavailable role.

13.3.1 SCI-FLT-MATCHED-A0-003-A: complete named-role covariance scope

Every authorized entry and correlation of the actual produced field is in scope on the full valid output domain for one explicitly named covariance role, with exact random variables, population, conditioning, units, rank/null, response, and generation. U1 conditional stochastic covariance is not a joint U1+U2 calibration/template product by implication.

13.3.2 SCI-FLT-MATCHED-A0-003-B: named projected scientific covariance scope

Only $P_c C_{U1, \text{realized}}^{(r)} P_c^\top$ for an explicitly named covariance role r and consumer projection is in scope. The role, projection basis, measure, rank, validity, and consumer are exact; all outside correlations remain unknown.

13.3.3 SCI-FLT-MATCHED-A0-003-C: covariance unavailable

Missing parent-covariance or cross-term authority is typed unavailability. The signal may remain complete, but no operational covariance, covariance-based standardization, significance, draw, independence, or covariance-dependent use follows. A separately theorem-supported $v_{\text{GLS}, \text{reference}}(p) = d_p^{-1}$

under SCI-FLT-MATCHED-A0-001-A may remain available. This typed companion status does not make the signal bundle partial unless an exact named-use policy requires the covariance companion.

13.3.4 SCI-FLT-MATCHED-A0-003-D: exact resident covariance representation

Represent the selected scope by an explicit array or exact structured object. The representation type and query vocabulary are declared; every supported entry/quadratic form is exact, and no representation-only change alters the scientific covariance identity.

13.3.5 SCI-FLT-MATCHED-A0-003-E: exact lineage/on-demand covariance representation

Persist exact immutable lineage sufficient to reconstruct every supported query without undocumented state. Each result is immutable and carries the same covariance identity. Missing lineage or unsupported queries are unavailable, never zero.

13.4 SCI-FLT-MATCHED-A0-004: Immutable-state representation

The scientific invariant for every alternative is exact reproducibility of the resolved immutable state and its required query vocabulary. Representation alone does not change the estimator or signal generation. After the scientific query vocabulary is fixed, A–C are engineering representation selections.

13.4.1 SCI-FLT-MATCHED-A0-004-A: full materialization

Persist every resolved state element with exact indexing, units, population, generation, and canonical identity. Reload shall reproduce every supported operator/state query exactly under the numerical profile.

13.4.2 SCI-FLT-MATCHED-A0-004-B: compact exact state

Persist an exact factorization or sufficient-state representation plus complete query semantics. Discarded information is allowed only when proved irrelevant to every supported query. Unsupported queries are unavailable.

13.4.3 SCI-FLT-MATCHED-A0-004-C: exact lineage reconstruction

Persist immutable inputs, Learn/Resolve algorithms and versions, environment facts, and identity needed for deterministic reconstruction. Missing or mutable lineage fails dependent routes; relearning is not a fallback.

13.5 SCI-FLT-MATCHED-A0-005: Response science and exact representation

Before representation, every route binds fixed/FP/realized/reference type, anchor domains, index/measure, query vocabulary, validity, units/calibration, state/operator/template generation, consumer scope, retention, and failure. FLT does not guess a future FRUIT query. After the scientific response type/domain/query vocabulary is fixed, A–C are engineering representation selections.

13.5.1 SCI-FLT-MATCHED-A0-005-A: full response materialization

Materialize every response entry in the authorized domain with exact index maps and identity. A matrix does not by itself authorize a beam, moment, integral, or other reduced interpretation.

13.5.2 SCI-FLT-MATCHED-A0-005-B: exact structured response

Persist an exact factorization or mathematical object that answers the entire declared query vocabulary without unproved invariance. Unsupported queries are typed unavailable.

13.5.3 SCI-FLT-MATCHED-A0-005-C: exact lineage/on-demand response

Persist complete immutable lineage needed to reconstruct supported response queries without undocumented state. Query results are immutable; failure does not substitute a reduced kernel.

13.6 SCI-FLT-MATCHED-A0-006: Role-separated SCI-VAL verdict representation

The normative policy is one dependency graph over seven separate verdicts: PA, SA, SP, CU, NU, RU, and FH. Each retains request, applicability, eligibility, realization, producer facts, policy, action, lifecycle, evidence, and version. The alternatives are lossless engineering representations of the same owner-disposed policy and do not require a scientific-owner selection.

13.6.1 SCI-FLT-MATCHED-A0-006-A: seven separate records

Persist one immutable record per role. A view may list all verdicts but has no actionable scalar aggregate.

13.6.2 SCI-FLT-MATCHED-A0-006-B: grouped layered records

Group roles into declared layers only while retaining seven independently addressable subverdicts and the full dependency graph. A layer summary cannot replace the subverdict vector.

13.6.3 SCI-FLT-MATCHED-A0-006-C: one seven-role verdict vector

Persist one composite with seven separately addressable complete records. Consumers request a named component. A scalar may be human-readable only and shall never drive action.

13.7 Option closure

Until exact dispositions exist, affected weighting, numerical-conformance, covariance, state-query, response-query, SCI-VAL, and handoff routes remain unavailable. Scientific-owner dispositions choose scientific meanings; engineering preregistration/representation declarations choose exact realization form only after those meanings are fixed; incomplete catch-alls trigger successor authorship. No disposition establishes implementation conformity, response or covariance fidelity, observational validation, performance, readiness, production, freeze, or Unity status.

14 Edge cases, exclusions, and failure routing

Condition	Scientific meaning	Required handling
Empty or incomplete application support	No defined reference estimand at the affected anchor	Mark unavailable before evaluation; no padding, extension, imputation, partial estimate, or renormalization.
Numerical fill near an edge	Merely a computational placeholder	Admit an output only after exact erosion proves zero dependency on every filled value.
Missing payload on a construction-only row with exact-zero c_{pq}	The row is in $\mathcal{D}_{\text{loc}}(p)$ but not $\mathcal{S}_{\text{apply}}(p)$	Do not dereference the payload; the amplitude remains defined and unchanged. If $c_{pq} \neq 0$, the row becomes an active dependency and the anchor is Unavailable. No coefficient threshold is permitted.
$d_p = 0$, $d_p < 0$, nonfinite d_p , or complex d_p	Template is null, weighting is invalid, or numerical identity failed	Typed unavailability or attempt failure; never publish zero amplitude, infinite variance, or fallback.
Template lies partly in $\mathcal{N}(W_p)$	Some declared template modes are unestimable	Allowed only if the selected option explicitly defines the same estimable subspace and $d_p > 0$; record response loss and null space. Otherwise unavailable.
Indefinite or non-self-adjoint W_p	Weight is outside the admitted scientific bilinear family semantics	Fail selected SCI-FLT-MATCHED-A0-001 closure unless an exact distinct operator and inner product were separately authored; no silent symmetrization.
Covariance zero-variance mode	Parent law declares deterministic variation in that direction	Keep separate from excluded, weighting-null, and template-unidentifiable modes. It is outside the v0.1 positive-definite-subspace theorem unless separately authored; do not broaden the stated optimum.
Excluded or infinite-variance mode	Policy deliberately removes a direction from the estimable subspace	Record the exact projector and response loss; do not call it a covariance zero mode.
Weighting-null mode	W_p assigns zero action to a direction	Record the weighting null space; no covariance meaning follows without separate authority.
Projected template is zero	The requested amplitude is unidentifiable in the declared subspace	Anchor unavailable; no Moore–Penrose or normalization fallback.

Condition	Scientific meaning	Required handling
Zero, negative, or non-finite field-power bin	No finite declared reciprocal weighting for that mode/bin	Apply only owner-declared regularization within the frozen state; otherwise exclude as a declared null mode or make route unavailable. Never infer infinite precision.
Sparse radial bin or large angular scatter	Radially averaged PSD lacks declared information or radial sufficiency	Report the preregistered effective-sample, anisotropy/dispersion, and leakage diagnostics. They are not v0.1 admission predicates and have no owner-set failure thresholds. Exact weight/denominator/support predicates still apply; no neighboring-bin interpolation unless separately authored.
Unknown covariance entry	Scientific knowledge is absent	Preserve unknown; do not encode zero, diagonal independence, or a usable inverse.
Available W but unavailable C_{parent}	Weighting exists without covariance authority	Amplitude filtering may proceed under a selected weaker option; optimality, reference variance, and operational covariance are unavailable.
Exact GLS premises with operational covariance unavailable	Reference theorem and operational product have different roles	Preserve $v_{\text{GLS,reference}} = d_p^{-1}$ when independently established; mark the operational covariance role Unavailable and forbid covariance-derived uses.
Available C_{parent} with mismatched population/state	Covariance does not describe this realization	Covariance route unavailable; no rescaling or population substitution.
Template rescaled by $c \neq 0$	Amplitude convention changes	Create a new template/realization identity. The $\hat{a} \mapsto \hat{a}/c$ law holds only for a pure amplitude-coordinate reparameterization with $\mathcal{D}_{\text{loc}}$, E_p , W_p , g , support, and non-template state fixed. Any re-Learn or re-Resolve operation is a new realization outside that law; units and lineage must follow in every case.
Template phase or anchor changed	Scientific response convention changes	New template and realization; never treat as a numeric implementation detail.

Condition	Scientific meaning	Required handling
Template mismatch to a real signal	Base matching-template estimand does not equal an unconstrained source property	Report only response-weighted shape amplitude; no bias correction, morphology, or new uncertainty without separate authority.
Parent beam or calibration exists	Provenance is present, not inherited matched-filter meaning	Preserve lineage; require exact authority before flux, matched beam/solid angle, extended-source, or calibration-covariance claim.
Coadd membership changes	Parent identity changes	New coadd parent, state, realization, response, and products; do not reuse observation/coadd state by implication.
NOI member fails K_{NOI} or suggests a new state	Admission or per-member relearning would change the ensemble method	Mark the member or affected anchor incompatible/unavailable before Apply. A learned update creates a successor generation; no support adaptation or result-dependent admission.
Approximation misses one null or support mode	Estimand/validity differs even if aggregate error is small	Fail the preregistered profile; if intentional, assign a distinct scientific identity. No percentile hides the disagreement.
Iteration/tail cap reached	Work limit, not convergence evidence	Accept only if every selected quantitative envelope bound independently passes; otherwise fail.
Optional companion fails after signal completion	Companion atomicity failed, not necessarily signal science	Mark companion failed; preserve signal complete publication candidate unless exact named-use policy made the companion required.
Optional role not requested	No child or companion was requested	Record not_requested ; do not narrow $\mathcal{V}_{\text{signal}}$ or block another role.
Normative per-anchor unavailability	The validity rule defined no amplitude at that anchor	Preserve the unavailable status inside an otherwise role-complete field; this does not by itself make the bundle partial.
Required signal bundle member fails	A required field, mask, identity, response reference, lifecycle record, or lineage member is absent	Fail the entire signal bundle; do not disguise the missing member as unavailable pixels inside an otherwise complete product.
Successor state or option is approved	Scientific identity advances	Create immutable successor generation and mark predecessor superseded only by policy; never mutate predecessor facts.

Condition	Scientific meaning	Required handling
Unselected or unavailable option requested	No authorized route exists	Fail closed and name the missing owner disposition; no automatic selector or fallback.
Response query is unsupported	Scientific interface does not define that quantity	Return typed unavailability; never approximate with a kernel, radial profile, parent beam, or zero.
Fixed-state response requested as full-procedure response	State-learning dependence has not been rerun	Return full-procedure response unavailable and retain the fixed-state type; no consumer may strengthen it.
Reference response or covariance attached to a realized field	Comparator is being mislabeled as the actual product consequence	Fail the companion or bundle policy; attach the operational realized response and named-role $C_{U1, \text{realized}}$ when authorized.
Failure-bearing operational covariance sample	$P_C F_g(m)$ is not a fixed-dimensional finite numerical random vector almost surely on the predeclared domain	Mark the covariance role unavailable . Do not condition on success, censor, pairwise-delete, or change domains without a separately named population and covariance role; signal status is unchanged unless named-use policy requires the companion.
FLT to FRUIT consumer asks outside $\mathcal{Q}_{\text{FLT}}^{0.1}$ or for FRUIT behavior	Request crosses the current envelope or package authority	Return typed unavailability and require a successor FLT envelope or separate FRUIT contract as applicable; existing FLT signal and handoff facts remain unchanged.
Filtered-map peak is examined	A numerical feature is not an authorized source object	No detection, candidate, significance, completeness, purity, fit, morphology, catalog, or SRC claim follows.
Historical title or filename contains “optimal”	Human-facing terminology is not theorem evidence	Require explicit <code>optimality_status</code> ; under SCI-FLT-MATCHED-A0-001-C it is always not_claimed .

14.1 Explicit method exclusions

Posterior/Wiener sky reconstruction, generic ordinary-convolution semantics, source or candidate learning, data-thresholded destriping, automatic selectors/fallbacks, adaptive edge/background behavior, the historical high-pass/delta case, partial-support estimation, FLT-owned coaddition, source analysis, and FRUIT algorithms are not authorized by this contract. A proved special-case numerical coincidence with normalized correlation/convolution retains the SCI-FLT-MATCHED estimand and does not authorize generic convolution semantics. Other methods require separate scientific contracts if later commissioned.

SCI-FLT-FIXED remains a separate protected package. This contract neither selects, modifies, supersedes, nor supplies scientific authority for that method.

The historical compatibility term “Wiener filter” may refer to this method only when accompanied by the exact SCI-FLT-MATCHED identity and may not imply a sky prior, posterior estimand, or Wiener reconstruction.

14.2 Nonclaim closure

Passing a scientific contract consequence or an engineering conformance test establishes only that named claim for the exact immutable realization. Validation-profile pass is not achieved performance. Achieved performance is not readiness. Readiness is not production. None of those layers can select an authored option, repair missing scientific authority, or broaden this validity domain.

Part III

Owner-disposition guide

15 What remains for the scientific owner

This draft separates scientific-owner decisions, engineering preregistration/representation selections, and successor-authorship triggers. The owner disposes only the remaining scientific meanings. Engineering selects exact representations and one numerical profile after those meanings are fixed. The decision ledger records the exact routing.

15.1 Human-title and optimality disposition

The package ID remains SCI-FLT-MATCHED. The recommended human title is “Matched-template map amplitude estimation.” The second acceptable title is “Normalized matched-template map filtering.” A third option retains the historical title but requires an explicit `optimality_status` on every product and boundary. In every option, “optimal” is reserved for SCI-FLT-MATCHED-A0-001-A with every theorem premise established; SCI-FLT-MATCHED-A0-001-C and every weaker route carry `not_claimed`. No title, filename, or ID is claim evidence. This r0.5 draft retains the historical title provisionally and does not select for the owner.

Family	Authority and decision	Route unavailable before disposition
SCI-FLT-MATCHED-A0-001	Scientific owner: concrete weighting authority for observation and coadd separately. B/D remain successor-authorship triggers until concrete W_p objects exist. The C-route diagnostics are mandatory evidence but are not v0.1 validity gates. Recommended: authorize A as the sole exact local-GLS optimal/reference-variance route; permit C separately as <code>radially_symmetrized_field_power_spectral_weighting</code> with no noise/covariance/stationarity/isotropy/optimality claim.	A weighting realization, any optimality label, and any variance meaning derived from the weight.
SCI-FLT-MATCHED-A0-002	Recommended: engineering pre-registers one exact-operator numerical profile or records typed unavailable. Scientific owner acts only to commission a concrete distinct successor operator.	Any numerical-conformance statement.
SCI-FLT-MATCHED-A0-003	Scientific owner: named covariance role and complete, projected, or unavailable scope. Every bundle retains role identity/status; unavailable scope does not invalidate the signal unless a named-use policy requires it. Engineering: resident or lineage representation afterward. Reference $v_{\text{GLS,reference}}$ remains separately typed.	Covariance publication, standardization, or covariance-dependent use.
SCI-FLT-MATCHED-A0-004	Scientific owner: finite required state-query vocabulary. Engineering: full, compact-exact, or lineage representation.	State audit/reanalysis and handoff queries requiring reconstructable state.
SCI-FLT-MATCHED-A0-005	Scientific owner: response type, domain, query vocabulary, validity, and consumer scope. Engineering: full, structured, or lineage representation.	Response-dependent use and FLT-to-FRUIT response queries.

Family	Authority and decision	Route unavailable before disposition
SCI-FLT-MATCHED-AO-006	Scientific semantics are decided: seven roles and dependency graph. Engineering chooses separate, grouped, or vector layout.	Registered SCI-VAL named-use evaluation still awaits its policies and facts.

No option choice alone authorizes a numerical route. After owner disposition, an implementation still needs separate conformity evidence; profiles need separate SCI-VAL evaluation; observational validation, measured performance, readiness, and production remain separate decisions.

A Author-packet basis and firewall statement

Stage B r0.1 used only these eight admitted objects:

- AUTHOR_PACKET_MANIFEST.md and SCOPE_BRIEF.md;
- AUTHOR_SUPERSESSION_COVER.md and AUTHOR_CONVENTIONS_AND_OWNERSHIP.md;
- SCIENTIFIC_OWNER_DECISIONS_FOR_AUTHORSHIP.md;
- AUTHOR_OPERATOR_STATE_AND_PRODUCT_TAXONOMY.md; and
- AUTHOR_BOUNDARIES.md and REQUIRED_AUTHORED_OPTION_SETS.md.

The r0.2 revision additionally used only the scientific owner’s exact targeted closure directive dated 2026-08-31. The r0.3 repair used the owner’s instruction to apply the independent ChatGPT Pro r0.2 document-review findings; the reviewer verified the two PDF hashes and inspected no implementation. The r0.4 repair used the second independent ChatGPT Pro review of the exact r0.3 PDF pair and the owner’s approval of selectable covariance scope plus diagnostics-only AO-001-C admission policy. The r0.5 repair used only the exact scientific-owner directive dated 2026-09-01 and its resulting owner-facing disposition records; it inspected no implementation. Links in the packet were not opened. No implementation, configuration, schema, test, audit, validation, reduction, generated product, repository history, neighboring package, external source, or model-memory source supplied scientific content.